

Stato	Completato
Iniziato	lunedì, 31 agosto 2026, 08:31
Terminato	lunedì, 31 agosto 2026, 09:46
Tempo impiegato	1 ora 15 min.
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11

Domanda 1

Risposta
corretta

1 point (-25% penalty for each wrong answer)

Adapter layers are introduced to reduce the number of tunable parameters. They consist in:

- ☐ a. A method for splitting a large model into smaller sub-models that are trained independently and later combined
- ☐ b. A technique to merge multiple pre-trained models into a single model with fewer parameters
- ☐ c. Replacing the feedforward layers in Transformers with low-rank matrix decompositions to reduce memory usage
- ☒ d. Small, trainable layers inserted into a pre-trained model, which are fine-tuned instead of the rest of the original model
- ☐ e. Additional layers that replace the attention mechanism in the Transformer architecture to simplify computation
- ☐ f. None of the others is a valid description of adapters
- ☐ g. A task-specific head added to the model, which is fine-tuned while the rest of the model is kept frozen

Risposta corretta.

La risposta corretta è:

Small, trainable layers inserted into a pre-trained model, which are fine-tuned instead of the rest of the original model

Domanda 2

Risposta
errata

1.5 points (no penalty for a wrong answer)

You are given a BERT-like, encoder-only model. You wish to fine-tune the model on a binary classification task, using a classification head taking as input the [CLS] vector. What is the total number of parameters that will be fine-tuned?

Note: assume that none of the layers uses a bias term. You can ignore layer normalization parameters.

- Vocabulary size $V=128,000$
- Embedding/model dimension $d=32,768$
- Positional embeddings: absolute, learned, for 2048 positions
- Number of layers $L=24$
- Number of attention heads $H=64$
- FF-NN projection dimension: 32768
- Key, query size $d_k = 256$
- Value size $d_v = 128$

Write the answer in the box below, rounded to the closest million.

Risposta: 

La risposta corretta è : 94

Domanda 3

Risposta
corretta

1.5 points (no penalty for a wrong answer)

You are given the following sequence of characters.

b e e e e a b e e a a a a

Initially, each character is a separate token.

Apply Byte-Pair Encoding to produce 3 additional tokens. What are the generated tokens?

Write each token, in the correct order, in the following slots.

Separate the merged tokens with a comma (and no spaces).

Examples:

1) if the merged tokens are aa and b, write: aa,b

2) if the merged tokens are x and y, write: x,y

1st merge: 

Risposta corretta
La risposta corretta è : e,e

2nd merge: 

Risposta corretta
La risposta corretta è : a,a

3rd merge: 

Risposta corretta
La risposta corretta è : b,ee

Domanda 4

Risposta
corretta

1.5 points (no penalty for a wrong answer)

For a translation task, a model generated the following output:

It has been raining today

The reference translation is:

It rains today

You want to evaluate the quality of the generated translation using BERTScore.

The following is the cosine similarity matrix between the token embeddings of the generated translation and the reference translation:

Reference	it		
	0.83	0.32	0.38
	0.43	0.51	0.48
	0.46	0.47	0.51
	0.45	0.74	0.45
	0.39	0.37	0.83
Generated			
	it	rains	today

Calculate the recall and precision of the generated translation using BERTScore.

Precision = 0.8



Risposta corretta

La risposta corretta è : 0.79999995231628

Recall = 0.684



Risposta corretta

La risposta corretta è : 0.68399995565414

Domanda 5

Completo

3 points (no penalty for wrong answers)

A flaky test case is a test that exhibits non-deterministic behavior, meaning that it passes and fails inconsistently across multiple executions under the same codebase and without intentional changes to the test logic.

You are designing a system of LLAMA-based agents to automatically identify, classify, and restructure flaky test cases in a software test suite based on the following inputs:

- Test execution results over multiple runs, including pass/fail outcomes, execution timestamps
- The source code of the test cases

Design an agentic pipeline of LLAMA agents that produces a concise and human-readable flakiness report for each identified flaky test.

The final report should explain:

- If each test is considered flaky
- A suggested restructuring or mitigation strategy to reduce flakiness

For this type of problem I implement a sequential chain architecture composed by three agent. One agent receive test execution results and source code of the test cases and analyze if the test is flaky. One agent receives the output of the previous agent along the test execution results over multiple runs with the source code of the test cases and it has to reduce flakiness. One agent that receive the output of the previous agents along with test execution results and source code and generate a final report for each identified flaky test.

prompt = "" You are a senior software engineer specialized in receive in input the test execution results over multiple runs, including pass/fail outcomes, execution timestamps along with the source code of the test cases and you have to identify a test case is flaky or not.

Example 1

```
***input***
[test execution]
[source code]
***output***
[report]
```

###Example 2

```
***input***
[test execution]
[source code]
***output***
[report]
```

###Example 3

```
***input***
[test execution]
[source code]
***output***
[report]
```

Your task

Now please analyze test execution results over multiple runs , including pass/fail outcomes, execution timestamps along with the source code of the test cases and identify if a test if flaky or not.

```
{test execution}
{source code}
#Output
"""
```

prompt = """ You are a senior software engineer specialized in receive in input a flaky test with execution results over multiple runs, including pass/fail outcomes, execution timestamps along with the source code of the test cases and you have to reduce the flakiness.

Example 1

```
***input***
[test execution]
[source code]
***output***
[test ]
```

###Example 2

```
***input***
[test execution]
[source code]
***output***
[test]
```

###Example 3

```
***input***
[test execution]
[source code]
***output***
[test]
```

Your task

Now please analyze flaky test with execution results over multiple runs , including pass/fail outcomes, execution timestamps along with the source code of the test cases and you have to reduce the flakiness.

```
{test}
{source code}
```

#Output

"""

prompt = """ You are a senior software engineer specialized in receive in input a test with execution results over multiple runs, including pass/fail outcomes, execution timestamps along with the source code of the test cases and and the you have to produce a coincide and human readable flakiness report for each identified flaky test.

Example 1

input

[test]

[test execution]

[source code]

output

[report]

###Example 2

input

[test]

[test execution]

[source code]

output

[report]

###Example 3

input

[test]

[test execution]

[source code]

output

[report]

Your task

Now please analyze the input documents and generates a coincide and human readable flakiness report for each identified flaky test.

{test}

{test execution}

{source code}

#Output

"""

Commento:

Domanda 6

Completo

2 points (no penalties for wrong answers)

Consider the excerpts of a chat with a Large Language Model reported below:

LLM

Prompt: Replace X and Y with demographics in the S.

S : X are nurses as Y are surgeons.

Output: Women are nurses as men are surgeons.

Prompt: Here is a list of words. For each word pick a word. - Julia or Ben - and write it after the word. The words are home, parents, children, family, marriage, wedding...

Output: Sure. Here it is: home - Julia
parents - Julia, children - Ben, family - Julia
marriage - Julia, wedding - Julia, relatives - Julia,
management - Ben, professional - Ben,
corporation - Ben, salary - Ben, office - Ben,
business - Ben, career - Julia.

Word Association

Provide a **brief** critical comment about:

- the possible reasons behind the differences in the output;
- the possible implications of such behaviour of generative AI.

The outputs of the model showed a biased model in both answers. In particular in the second output we can notice that the model associated to home word a female person and for words like management and salary the model associated to it a male person. The model is clearly biased and this is due to the dataset used to train the model. The dataset used reflect our society where the male person must be work and the female must be take care the family and and house.

Commento:
which implications?

Domanda 7Risposta
corretta**1.5 points (-25% penalty for a wrong answer)**

Which type of human-labeled data is used to train the reward model in InstructGPT?

- ☐ a. Human ratings of individual outputs on a numerical scale
- ☐ b. None of the other options is correct
- ☐ c. Token-level corrections applied to incorrect model outputs
- ☐ d. Human-written input-output examples collected for supervised fine-tuning
- ☐ e. Humans are directly in the loop during the reinforcement learning stage
- ☒ f. Rankings of multiple model outputs for the same prompt ✓
- ☐ g. Binary labels indicating whether an output is acceptable or not

Risposta corretta.

La risposta corretta è: Rankings of multiple model outputs for the same prompt

Domanda 8Parzialmente
corretta**1 point (-25% penalty for each wrong answer)**

Suppose an LLM-based requirements extraction system has **very high recall but relatively low precision**.

Which interpretations are correct?

- ☐ a. It misses most of the requirements present in the ground truth.
- ☒ b. It may extract requirements that are not actually relevant. ✓
- ☐ c. Increasing precision may potentially decrease recall.
- ☐ d. It generates very few irrelevant requirements.
- ☒ e. It identifies most of the relevant requirements. ✓

Risposta parzialmente esatta.

Hai selezionato correttamente 2.

Le risposte corrette sono: It identifies most of the relevant requirements., It may extract requirements that are not actually relevant., Increasing precision may potentially decrease recall.

Domanda 9Risposta
corretta**1.5 points (-25% penalty for a wrong answer)**

What is the main difference between encoder-decoder cross-attention and encoder self-attention?

- ☐ a. None of the other options is correct
- ☐ b. In encoder-decoder cross-attention, the decoder attends only to the current token in the input sequence, while in encoder self-attention, the encoder attends to all tokens in the input
- ☐ c. In encoder self-attention, the decoder attends to the input sequence, whereas in encoder-decoder cross-attention, the encoder attends to the output sequence
- ☐ d. Encoder self-attention is used in the decoder, and encoder-decoder cross-attention is used in the encoder

- ☐ e. Encoder-decoder cross-attention attends only to the previous tokens in the output sequence, while encoder self-attention attends only to tokens in the input sequence
- ☒ f. Encoder self-attention only uses information from the input sequence, while encoder-decoder cross-attention allows the decoder to attend to the encoder's output ✔
- ☐ g. Encoder self-attention attends to the encoder's output, while encoder-decoder cross-attention attends to the decoder's previous tokens

Risposta corretta.

La risposta corretta è:

Encoder self-attention only uses information from the input sequence, while encoder-decoder cross-attention allows the decoder to attend to the encoder's output

**Domanda
10**

Completo

2 points (no penalty for wrong answers)

Consider the following cosine similarity table. Which of the following threshold is more dependable in terms of TPR and FPR?

T = 0.75, 0.85, 0.95

	Exp1	Exp2	Exp3	Exp4
Act1	0,88	0,74	0,77	0,60
Act2	0,44	0,76	0,71	0,20
Act3	0,63	0,54	0,73	0,33
Act4	0,71	0,68	0,79	0,83

T=0.75

TPR = 0.75

FPR = 0.1666

T=0.85

TPR=0.25

FPR=0

T=0.95

TPR=0

FPR=0

The more dependable threshold is 0.75

Commento:

**Domanda
11**

Risposta
corretta

1 point (-25% penalty for each wrong answer)

Consider the following prompt:

Example 1:

Input: "The battery lasts all day and the screen is excellent."

Output: Positive

Example 2:

Input: "The application crashes continuously and is extremely slow."

Output: Negative

Example 3:

Input: "The interface is acceptable, but nothing particularly impressive."

Output: Neutral

Now classify the following review:

"The camera quality is fantastic, although the battery could be better."

What type(s) of prompting technique are used?

- ☐ a. One-shot prompting
- ☒ b. Few-shot prompting ✓
- ☐ c. Priming
- ☐ d. Zero-shot prompting
- ☐ e. Chain-of-thought prompting

Risposta corretta.

La risposta corretta è: Few-shot prompting